Designing AI-Supported Psychotherapy Practice: A Research Through Design Study of Virtual Patients and Feedback

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Abstract

Introduction The growing demand for psychotherapy professionals has intensified the need for scalable and effective training methods. While recent advances in large language models offer promising opportunities to address two key challenges in psychology education: the limited availability of individualized feedback and the creation of virtual patient scenarios, existing research has largely focused on physiological and clinical contexts, leaving psychology education comparatively underexplored. As a result, it remains unclear where and how AI tools could appropriately support psychotherapy training.

Research Question How can AI-generated feedback support the training of psychology students in Motivational Interviewing?

Method A research-through-design approach was employed. Semi-structured interviews with a psychotherapy professor from the psychology department at SU were conducted to explore current training practices and challenges. Findings from these interviews informed the iterative development and evaluation of design concepts aimed at identifying meaningful roles for AI in psychotherapy training and ultimately a prototype feedback tool was tested with psychology students.

Results Students responded positively to the tool, highlighting its portability and the opportunity for reflective practice, though they consistently positioned it as a complement to existing methods rather than a replacement. Feedback design emerged as a particularly complex dimension, requiring a balance between timing, cognitive load, and qualitative and quantitative forms.

Discussion LLM-powered virtual patients show promise as a supplementary tool for psychotherapy training, though adaptive feedback approaches

may be needed to accommodate learners at different stages of development. Important ethical considerations remain, and future research should prioritize larger-scale evaluations to move beyond perceived usefulness toward measurable learning benefits.

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
VP	Virtual Patient
SP	Standardized Patient
MI	Motivational Interview
RtD	Research Through Design
AIED	Artificial Intelligence in Education

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1

Introduction

According to the World Health Organization (WHO), more than a billion people live with a mental health condition [26]. As a result, rising prevalence and awareness have increased demand for psychotherapy and mental health treatment[40] and mental health treatment. Therefore, demand for training psychotherapy professionals has also increased, making it one of the careers with a strong future[7]. Effective psychotherapy training requires not only theoretical knowledge but also supervised practice, which poses ethical risks if done with real patients[15].

The use of virtual patients (VP) appear to be an interesting tool for supporting and training new professionals, and has been recognized as a valuable pedagogical tool in health education, enabling students to engage in simulated clinical interactions in a safe environment. More specifically, VP are described as "*an interactive computer simulation of real-life clinical scenarios for the purpose of healthcare and medical training, education or assessment*"[13]. In psychology education, VP use can support the development of skills such as interviewing, reflective listening[43], and empathy, which are particularly central to approaches such as Motivational Interviewing (MI)[24]. MI is a technique specifically developed to help motivate ambivalent patients to change their behavior[4] and it is particularly challenging to learn as it requires practitioners to master a subtle conversational style that balances empathy, reflective listening, and guidance without disregarding the patient's autonomy. MI serves as the initial training context for this thesis. However, conventional VP systems are often resource-intensive to develop and difficult to adapt dynamically, limiting

their scalability and their ability to represent the open-ended nature of psychotherapy interactions [13].

Recent advances in AI offer promising opportunities to address these limitations. Beyond improving conversational flexibility, AI-driven systems may enable increasingly immersive and adaptive training environments. For example, Hamilton et al. describe how advancements in AI could allow healthcare students to experience "*fully immersive settings, complete with haptic feedback and AI-driven dialogue, allowing for more lifelike diagnostic and procedural experiences*" [19]. Such developments may be particularly valuable in psychotherapy training contexts, where nuanced interpersonal communication and adaptability are central components of clinical competence.

Emerging research has already begun exploring these possibilities in both clinical education[44] and psychology education through AI-generated therapy scenarios and simulated patient dialogues [17, 34]. While these studies demonstrate the potential of AI-generated VP to provide accessible and repeatable practice environments, they also reveal important limitations [8, 31]. In particular, providing timely and individualized feedback remains constrained by instructor availability. The use of LLMs may then help address not only the generation and adaptation of VPs, but also help provide individualized feedback [11].

Although AI-generated feedback has shown promise across multiple educational domains and is beginning to extend to therapeutic soft-skill development [22], the field still needs technologies that specifically support therapist training and feedback [36]. Implementation approaches remain domain-specific [21], and it remains unclear how psychotherapy professionals would leverage such tools within psychology education. Existing work exploring LLMs in motivational interviewing contexts has largely focused on the model acting as the therapist [43, 1], rather than as a VP supporting trainee practice, leaving this configuration comparatively underexplored.

1.1 Problem Statement

Despite growing interest in AI-supported educational tools, it remains unclear how LLM-based systems can be meaningfully integrated into psychotherapy training, particularly in the context of VP scenarios for MI practice. While VP are widely used to support simulated clinical interaction, existing systems often struggle to provide scalable, adaptive, and timely

feedback that aligns with the needs of skill development in open-ended therapeutic communication.

Addressing this uncertainty requires an exploratory approach that engages both psychotherapy professionals and psychotherapy students in exploring potential AI-supported training practices.

To address this gap, this thesis adopts a research-through-design (RtD) approach [37], exploring how AI could support psychotherapy professionals in training students. Rather than assuming a fixed solution, the research uses iterative design and reflection to identify potentially meaningful roles for AI within psychology education.

1.2 Research Question

The research is guided by the following primary research question:

How can AI-generated feedback support the training of psychology students in Motivational Interviewing?

This question is further supported by the following secondary research questions:

- What challenges and unmet needs do psychotherapy professionals identify in current Motivational Interviewing training practices?
- What challenges and unmet needs do psychology students experience in current MI training practices?
- What design considerations emerge when developing AI-generated feedback systems for Motivational Interviewing education?

1.3 Delimitation

It is important to note that this thesis focuses on psychotherapy training in an educational context and does not address the use of AI in real clinical decision-making or patient treatment. The research emphasizes the perspectives of psychotherapy professionals and the design of training-support tools, rather than developing fully deployed or clinically validated AI systems.

2

Extended Background

2.1 Theoretical Framework

2.1.1 Psychology Education and Clinical Skills Training

A central concept in both medical and psychology education is the development of clinical reasoning skills. Clinical reasoning has been defined as "*including the physician's integration of her own (biomedical and clinical) knowledge with initial patient information to form a case representation of the problem*" [18]. Effectively teaching and practicing this skill with healthcare students has long been a challenge in educational settings. Traditionally, universities relied on real patients for training purposes; however, ethical concerns gradually led to the replacement of real patients with Standardized Patients (SP) [2].

SP encompass "*both actual patients trained to present their illness in a standardized way and simulated patients without the illness but trained to simulate it in a standardized way*" [19]. Since their introduction in 1970, SP have become a widely adopted gold standard in medical education, with over 180 universities across the United States incorporating SP-based training into their curricula [16]. Despite their benefits, the use of SP is associated with notable limitations. From an economic perspective, employing SP incurs substantial costs for universities. In practice, reliance on human actors introduces variability and potential errors into training outcomes [38]. Furhtermore, certain demographics are underrepresented in SP scenarios; for example, children and adolescents are less frequently included, which may limit the diversity of clinical exposure[30].

Virtual Patients

VP were first introduced in 1971 as a cost-effective alternative to SP [8]. Although their adoption was initially limited, the COVID-19 pandemic and the rapid expansion of telehealth practices accelerated their implementation in health education [19].

VP systems vary widely in technological sophistication and pedagogical design. Low-fidelity VP typically employ decision-tree architectures or rule-based systems, providing structured interactions with limited adaptability. These systems are commonly used to support early-stage learning and the development of foundational clinical reasoning skills. High-fidelity VP, in contrast, often integrate AI, natural language processing (NLP), and speech recognition technologies, enabling more dynamic and context-sensitive interactions. Such systems aim to enhance realism and immersion, thereby supporting advanced clinical skill development [33], including not only diagnostic reasoning but also communication-based competencies and soft skills such as interviewing, empathy, and patient interaction.

The implementation of VP is widely different across institutions. In some contexts, VP have been introduced as partial or complete substitutes for SP, whereas in others they complement or replace traditional paper-based case studies. A frequently cited advantage of VP is their ability to reduce curricular constraints, as students can engage asynchronously and independently with cases, increasing flexibility and allowing learners to progress at their own pace [19].

Another key advantage of VP over SP is their capacity to simulate longitudinal clinical trajectories [19]. Learners can follow the progression of a case over time, whether modeling the evolution of a chronic medical condition or the development of a psychological intervention across multiple sessions. This longitudinal functionality allows students to observe dynamic changes in symptoms, treatment responses, and therapeutic processes—experiences that are difficult to replicate in single-session SP encounters.

Despite the aforementioned advantages, the implementation of VP has not been without controversy. Hamilton et al note "*One early concern in the VP literature was that students would not respond correctly or emotionally to the VP*"[19]. In contrast, Bearman et al. found that well-constructed computer-based interactions can elicit significant emotional engagement among medical students [3]. Taken together, these findings suggest that VP provide structured, repeatable, and emotionally engaging environments for clinical reasoning skills development[12][19]. However, the creation

and maintenance of such cases (particularly when longitudinal progression or narrative complexity is involved) can be resource-intensive[8]. This limitation raises important questions regarding scalability and adaptability, particularly in psychology education contexts where nuanced conversational dynamics are central. These constraints motivate the exploration of generative AI systems as a potential means of dynamically producing and supporting VP-based training experiences.

2.1.2 Motivational Interviewing

MI is a client-centered approach in psychotherapy that helps individuals explore and resolve ambivalence about behavior change. Rather than directing or persuading, MI encourages clients to articulate their own reasons for change, fostering intrinsic motivation. Something important to highlight is that MI "*requires the clinician to engage with the client as an equal partner and refrain from unsolicited advice, confronting, instructing, directing, or warning. It is not a way to "get people to change" or a set of techniques to impose on the conversation.*"[24]. The approach is guided by a collaborative, empathetic stance that emphasizes respect for the client's autonomy and personal values. Core MI principles include:

- Open-ended questions (as opposed to closed questions)
- Reflective listening/Reflections (simple, complex, double-sided, summaries)
- Affirmations
- Autonomy (as opposed to being argumentative or judging)
- Cooperation (as opposed to assuming an expert role)

MI is therefore widely used in contexts where clients may feel uncertain, resistant, or ambivalent, such as addiction, health behavior change, and mental health treatment, and it can be combined effectively with other therapeutic modalities like Cognitive Behavioral Therapy or Acceptance and Commitment Therapy.

Although MI includes structured communication principles and techniques, effective practice requires practitioners to continuously adapt their responses to the client's language, level of motivation, and expressions of ambivalence throughout the conversation. This makes MI particularly difficult to master, as clinicians must balance guidance while not being overly

directive. As a result, MI training commonly emphasizes repeated practice, observation, and individualized feedback on conversational performance rather than theoretical instruction alone [24]. Competence is often assessed using coding frameworks such as the Motivational Interviewing Treatment Integrity (MITI) code, which evaluates adherence to the core MI principles listed above and communication behaviors.

These characteristics make MI particularly dependent on opportunities for repeated practice and individualized feedback, creating a potential role for AI-supported systems capable of simulating and evaluating therapeutic interactions at scale.

2.1.3 Large Language Models and Prompt Engineering

LLMs are deep learning systems trained on large corpora of text, capable of generating contextually coherent and semantically rich language across a wide range of tasks [6]. Their capacity for open-ended dialogue, instruction following, and nuanced language generation has positioned them as particularly promising tools in educational and clinical simulation contexts [10], since they can offer immediate feedback at reduced costs[34].

Chen et al.[10] identifies five approaches to LLM deployment: re-training, fine-tuning, knowledge embedding, retrieval-augmented generation (RAG), and prompt engineering. The choice among these approaches depends on the availability of domain-specific data, computational resources, and the degree of behavioral control required. In educational contexts with limited labeled data and a need for rapid iteration, prompt engineering has emerged as the most accessible and commonly used approach [31].

Prompt engineering refers to the deliberate design of input instructions provided to an LLM in order to guide its outputs toward a desired behavior or format. Rather than modifying model weights, prompt engineering shapes model behavior at inference time. Common prompt engineering strategies include zero-shot prompting, few-shot prompting, chain-of-thought prompting, and persona-based prompting. Zero-shot prompting involves instructing the model to perform a task without providing examples, relying solely on the model's prior training and the clarity of the instruction. Few-shot prompting extends this approach by supplying example inputs and outputs that demonstrate the desired behavior or response format[6]. Chain-of-thought prompting encourages the model to generate intermediate reasoning steps before producing a final response, which has been shown to improve performance on complex reasoning tasks [41]. Persona-based prompting guides model behavior by assigning a specific

role or communication style to the LLM[32], such as instructing the model to act as a therapist, patient, or educator. This approach has become particularly relevant in conversational systems and VP simulations, where maintaining consistency and believable role behavior is essential.

It's important to note that effective prompt development follows an iterative process in which candidate prompts are tested, evaluated, and refined based on output quality [31]. This iterative nature introduces challenges for reproducibility and systematic evaluation, as prompt performance may vary across tasks and context.

In the context of AI-supported learning systems, these prompting strategies have been used to guide LLMs in generating feedback, adopting specific personas, and maintaining role consistency across extended interactions. The capacity of LLMs to sustain coherent, contextually appropriate behavior across multi-turn dialogues makes them particularly suited for VP simulations, where realistic and responsive interactions are essential. However, this also introduces risks related to output reliability and consistency, reinforcing the importance of careful prompt design and, where possible, human oversight of model outputs [31, 19].

2.2 Related work

2.2.1 AI use for VP Creation in Clinical Training

Several studies have explored the use of AI-generated VP in clinical training contexts [14, 44]. While these approaches demonstrate the potential of AI to simulate patient interactions, important distinctions exist between general medical training and psychology-focused training. In many medical domains, the symptoms presented to students are relatively well-defined and diagnostically bounded, which often enables more standardized interpretation and assessment.

In contrast, psychological assessment and psychotherapy are characterized by higher levels of ambiguity, subjective experience, and contextual complexity. Client presentations often depend on nuanced interpersonal dynamics, non-verbal communication, evolving narratives, and subtle linguistic cues[9], all of which may be difficult to capture in text-based AI-generated simulations[27]. These differences introduce unique challenges for the design and implementation of VP in psychological training contexts.

Furthermore, the focus of this thesis is not on supporting students' diagnostic accuracy, but rather on the development of therapeutic skills and

treatment-oriented communication. As a result, the requirements for VP systems extend beyond the accurate representation of symptoms to include the modeling of more complex human-like behaviors such as expressing ambivalence or resistance, and the provision of immediate feedback on the quality of student interactions.

2.2.2 AI use for VP Creation in Psychology Training

The use of VP in psychology training remains comparatively less established than in other clinical disciplines, such as nursing [34], which in part explains why the application of AI-generated VP in this domain is similarly underexplored relative to clinical contexts. Although prior research has examined the use of generative AI in VP development, existing studies have largely focused on physiological cases, leaving psychotherapy and psychology education as a notable gap in the literature.

One exception is the work of García Torres et al. [17], who investigated both the implementation of VP and the integration of generative AI within a psychology training context. Their findings revealed a discrepancy between perceived and measured outcomes: while students reported improvements in diagnostic ability following exposure to VP scenarios, objective assessment results did not reflect a corresponding increase in academic performance when compared to students who had not engaged with such scenarios. Comparable patterns emerged in a study by Sanz et al. [34], in which students interacted with ChatGPT as a VP and reported increased confidence and a greater sense of preparedness for clinical settings, particularly highlighting the "safe environment" provided by practicing with an LLM rather than a human interlocutor.

In the context of VP, Hamilton et al. [19] emphasize AI's "facility for developing sustained storylines," which enables students to interact with the same VP at multiple points in the course of an illness or to explore alternative trajectories. Such capabilities suggest that LLMs could support longitudinal, adaptive VP scenarios in psychology education. Importantly, the provision of automated feedback has been identified as a key aspect for future exploration, indicating a direct link between LLM-enhanced VP and the next focus of this thesis [17].

An important early example that partially reflects this direction is ClientBot [39], a conversational agent designed to simulate a patient while providing real-time feedback on counseling skill use. This dual-role configuration aligns with the emerging need for systems that combine interactional practice with automated feedback in psychotherapy training contexts. However, ClientBot was not evaluated with psychotherapy-related participants and relied on LSTM-based neural networks rather than LLMs, limiting its capacity for flexible, open-ended dialogue and adaptive feedback generation in complex therapeutic interactions.

2.2.3 Artificial Intelligence in Education and Feedback Systems

Stamper et al. define Artificial Intelligence in Education (AIED) as "*the intersection of technology, education, and psychology, placing a strong emphasis on supporting learners' needs with compassion and understanding*" [23]. Within this context, feedback constitutes a fundamental component of effective learning. Given that the primary objective of VP is to support the development of clinical skills through simulated interactions, the availability of meaningful feedback is essential for enabling students to reflect on and improve their performance.

AI could be able to bridge this gap by enabling the provision of immediate and individualized feedback following student interactions. Unlike traditional instructional settings, where feedback is constrained by instructor availability, AI systems can provide scalable and real-time responses to learner actions. Several studies have suggested that students are often unable to distinguish between feedback generated by instructors and feedback generated by AI systems [31, 35], and generally perceive AI-generated feedback as helpful [12]. These studies span multiple domains, including education sciences, mathematics, programming, and language learning.

More recent work has extended these findings to areas that require complex interpersonal and communication skills. For example, studies on LLM-based feedback systems in therapeutic training contexts [22] show that AI can provide structured and explainable feedback on therapist's responses, grounded in domain-specific criteria. Such systems have shown to not only support reflection but also lead to measurable improvements in performance, particularly in handling nuanced interactional challenges such as client resistance. This suggests that AI-generated feedback can move beyond general educational support and play a meaningful role in the development of soft skills. These findings are particularly relevant for VP systems, where the ability to assess and guide interaction quality is critical

for effective learning. Recent work has also demonstrated that LLMs can serve as effective reference-free evaluators for text generation tasks, where the model is prompted to generate a numerical quality assessment directly from the output[42], which is particularly relevant for therapeutic training contexts, where reference responses are difficult to define and evaluation must account for nuanced communicative qualities rather than surface-level text similarity.

Despite these promising findings, concerns remain regarding the reliability of LLM-generated feedback. Rudian et al. emphasize that such systems should operate within a human-in-the-loop framework to mitigate potential issues such as contextual inaccuracies and bias in model outputs [31]. Similar concerns are raised by Hamilton et al., who question how a productive coaching relationship can be maintained among instructors, students, and AI-based or VP systems when these technologies are integrated into educational settings [19].

In terms of implementation, Qinjin et al. [21] identify two primary approaches to feedback generation: data-driven methods, such as fine-tuning or pre-training models on prior student work and instructor feedback, and prompt-based methods, which rely on carefully designed prompts to guide model outputs. The authors do not report a clear difference in feedback accuracy between these approaches, suggesting instead that the choice of method should depend on factors such as data availability and the type of feedback required.

These findings suggest that AI-generated feedback can be effective to learners across a range of educational domains[31, 35], and that its application is beginning to extend to soft-skill development in therapeutic contexts [22]. However, implementation approaches remain domain-specific [21], and it remains unclear how domain experts such as psychotherapy professionals would leverage prompt-based methods to design meaningful feedback systems for VP-based training in psychology education, particularly in the context of MI.

3

Method

3.1 Research Strategy

This thesis adopts a research-through-design (RtD) strategy to investigate how AI-generated feedback can support the training of psychology students in MI. RtD is a design research approach in which knowledge is generated through iterative design exploration and critical reflection on design activities and artifacts. Rather than aiming to validate a predefined solution, RtD is particularly suited for more exploratory work, where requirements and constraints are not yet fully defined[37].

A key scoping decision made early in the research process concerned the identification of the primary end user. As the project was initiated in collaboration with the psychology department, it was necessary to establish whether the system was intended primarily to reduce instructor workload or to expand learning opportunities for students. Following initial consultations with the course instructor, the latter was adopted as the guiding objective, with instructor expertise informing the design process in a collaborative capacity.

The choice of RtD is also motivated by how this area has been studied so far. Most existing work follows mostly evaluative research designs, where a specific system is built and then assessed using predefined measures or structured study setups. For instance, García Torres et al.[17] use a cross-sectional design with standardized questionnaires, while Sanz et al.[34] combine quantitative measures with qualitative follow-up to explain

their findings. Fąferek et al.[14] similarly apply a multi-step evaluation approach involving expert review and student testing across different institutions.

These kinds of studies work well when the problem is already clear; when there are established ideas about what the system should do and how success should be measured. In the case of AI-supported MI training, however, many of these assumptions are still undefined. It is still unclear what role AI should take in VP systems, what meaningful feedback should look like, and how such tools should fit into existing teaching practices. RtD fits this kind of situation better because it allows the design space itself to be explored through iteration and reflection, rather than starting from a fixed solution. This makes RtD particularly appropriate for the present study, as the aim was to explore what an appropriate AI-supported training solution could look like rather than evaluate a predefined system.

3.1.1 Alternative Research approach

Another possible methodological pathway would have been a Design Science Research (DSR) approach[28], focused on the development and validation of a fully specified artifact. However, the objective of this thesis is not to produce and rigorously evaluate a finalized system, but rather to explore how AI might be meaningfully integrated into psychology education through iterative design. For this reason, a research-through-design strategy was deemed more appropriate, as it allows for exploratory, iterative inquiry into emerging design possibilities prior to large-scale implementation or formal evaluation.

3.2 Data Collection Methods

Data collection followed an iterative RtD process that combined an exploratory analysis of an existing virtual patient platform, semi-structured interviews (see Appendix 5.5), design-oriented work sessions with a psychology professor, and a later evaluation session with students. The use of semi-structured interviews allowed for a balance between consistency and flexibility, enabling the researcher to address predefined themes while also probing emerging insights in depth. This approach was particularly appropriate given the exploratory nature of the study and its focus on understanding experiences, pedagogical practices, and perceptions related to AI integration.

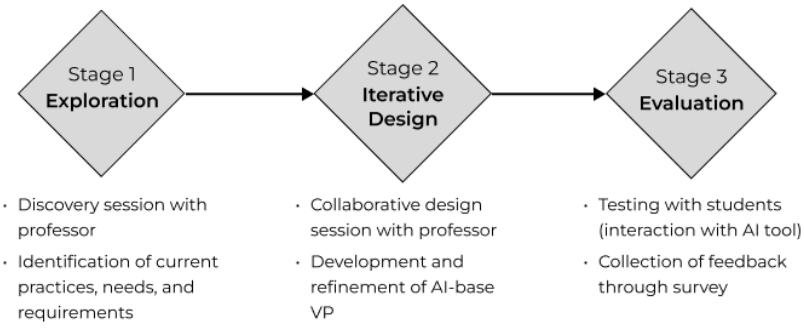


Figure 3.1: Data Collection Methodology

3.2.1 Phase 1: Exploration

The exploration phase began with an analysis of the existing Virtual Clinical Simulation (VCS) platform to understand how virtual patient-based clinical training was currently implemented and to identify initial design opportunities and limitations. The observations from this analysis informed the first semi-structured interview and design session with the psychotherapy instructor.

The initial session functioned as a discovery phase aimed at understanding existing teaching practices, identifying pain points, and exploring potential opportunities for integrating AI into the classroom. This session was intentionally open-ended, allowing the participant to articulate their perspectives, experiences, and priorities in depth. Given the instructor's pedagogical expertise and experience with MI (read more in Chapter 2), this phase provided a theoretical and practical foundation for subsequent design decisions.

Prior to the second session, an early prototype was developed to support more concrete discussion and reflection. This prototype was intentionally low-fidelity and served as a starting point for iterative refinement rather than a finalized concept.

3.2.2 Phase 2: Iterative Design

Following this exploratory phase, a second session took the form of a collaborative design discussion. During this session, early design ideas and illustrative AI-supported interaction concepts were introduced to facilitate

reflection and dialogue. This work session enabled the researcher to test emerging assumptions, explore alternative design directions, and identify ethical, pedagogical, and usability considerations related to the use of AI in educational contexts. The iterative and collaborative nature of this session aligns with design-oriented research approaches, where knowledge is generated through cycles of prototyping and reflection.

Prompt Design

As part of this process, LLMs were used to support the generation of VP scenarios and interaction prompts. The primary system used was ChatGPT (version 5.3), selected after exploratory testing alongside Microsoft Copilot. ChatGPT was ultimately chosen due to its more stable performance in role-based scenarios and its anticipated familiarity among students. Initial prompt development was iterative, with early versions generated using ChatGPT and refined through interaction with the system and in consultation with the instructor during the design sessions. Prompt development followed a minimal approach, whereby the initial prompt was kept as concise as possible and additional instructions were introduced only when specific failure modes were observed during testing [25], such as the system breaking character, providing feedback in an incorrect format, or defaulting to directive rather than MI-consistent language. It is important to note that the purpose of this process was not to systematically compare the performance of different LLMs, but rather to iteratively refine a prompt-based VP and feedback system within the practical constraints of the design process.

The evaluation of prompt outputs followed a two-phase approach inspired by Wu et al. [43], who propose separating evaluation into a non-expert phase focused on coherence and consistency, and an expert phase focused on domain-specific appropriateness. In the first phase, the researcher acted as a non-expert evaluator, assessing VP outputs for internal consistency, character breaks, and hallucinations. In the second phase, the MI trainer reviewed the outputs as a domain expert, evaluating their psychological and pedagogical appropriateness with respect to MI principles.

3.2.3 Phase 3: Evaluation

Evaluation in this thesis is qualitative and exploratory in nature. It focuses on assessing the perceived usefulness, realism, and pedagogical value of

the proposed AI-supported VP through user interaction rather than measuring technical performance or learning outcomes.

A third session was conducted with students, where the developed concepts were presented and evaluated. During this session, participants interacted with the proposed ideas and provided feedback through a structured survey (see Appendix 1), allowing for the collection of qualitative reflections on the proposed tool. The survey was selected over methods such as focus groups in order to encourage more independent and honest reflections from participants. Focus group discussions can be influenced by group dynamics, where more outspoken participants may shape the conversation and discourage disagreement or critical responses from others.

Students were intentionally introduced at this later stage of the study. Although they represent the primary end users of the proposed system, their limited prior exposure to MI at the time of data collection would have constrained their ability to meaningfully contribute to early-stage problem framing and pedagogical design. In accordance to this, student involvement was positioned as an early-stage prototype evaluation rather than a final assessment of a completed system; aimed at gathering preliminary user feedback, identifying usability issues, and exploring how the concepts were perceived by learners.

In addition, feedback from the psychology professor was incorporated throughout the iterative design process to ensure that the proposed approach aligned with existing teaching practices and educational values. Together, these perspectives informed an assessment of the methodological and practical relevance of the proposed design.

While iterative refinement based on user feedback is a central principle of design-oriented research, the limited duration of the study restricted the possibility of conducting additional evaluation cycles with updated prototypes. This constraint is acknowledged as a limitation of the study.

3.2.4 Participants

Stages 1 and 2

The initial stages of the study involved a series of interviews and collaborative design sessions conducted with the current trainer of the MI course. Access to an expert directly responsible for MI instruction provided a unique opportunity to ground the research in authentic pedagogical practice, ensuring that the design decisions were informed by first-hand experience

in both the therapeutic approach as well as the pedagogical aspect of the course.

Stage 3

The evaluation session involved 35 third-year students enrolled in the bachelor's degree in psychology. Students first interacted with the prototype in the presence of the researcher and were subsequently encouraged to continue practicing independently in their own time. Following this, participants were invited to complete a ten-question survey designed to capture their perceptions and opinions of the tool. Four students completed the survey; however, additional informal feedback was also provided verbally to the course instructor and taken into account in the overall evaluation.

3.2.5 Alternative Method

An alternative approach to the research question would have been a quantitative experimental study or a survey-based investigation aimed at measuring the effects of AI-supported tools on student learning outcomes or attitudes. Such approaches can produce generalizable results, but they typically require a clearly defined intervention and well-established evaluation metrics.

Given that the role of AI and LLMs in psychology education is still emerging, this study did not aim to assess predefined outcomes. Instead, the research focused on exploring and shaping potential design directions through iterative, qualitative inquiry. While a survey was employed in the later stage of the study, it was used as an exploratory tool to gather student feedback on proposed concepts rather than to generate statistically generalizable findings.

Nevertheless, quantitative and outcome-based approaches may represent valuable directions for future research once the conceptual and methodological foundations of AI-supported learning in this domain have been more clearly established.

3.3 Data Analysis Methods

The preliminary analysis of the VCS platform was conducted through a design-oriented artifact analysis. The platform was examined to identify

both pedagogical strengths and limitations relevant to MI training, with observations serving as an initial source of design inspiration and discussion during the expert elicitation sessions [29].

The interviews and collaborative design sessions with the MI trainer were treated as an expert elicitation process following Hoffman et al. [20]. Instead of applying a formal thematic analysis[5], the objective was to elicit domain-specific knowledge regarding current pedagogical practices, unmet training needs, and design considerations for integrating AI into MI training. Interview recordings were transcribed using Notta software and subsequently reviewed alongside notes from the design sessions. The elicited knowledge was synthesized into design considerations that informed successive iterations of the prototype.

Student feedback was analyzed qualitatively to identify recurring perceptions regarding the usefulness and realism of the proposed AI-supported VP. Given the exploratory nature of the evaluation and the small number of participants, responses were interpreted descriptively rather than subjected to formal qualitative coding. Particular attention was paid to areas of agreement and disagreement among participants, as well as suggestions for improvement that could inform future iterations of the prototype.

Across these activities, the analysis focused on generating design knowledge rather than producing generalizable empirical findings, with insights from each stage informing subsequent design decisions.

3.4 Ethical considerations

Participants were informed about the purpose of the study, the recording of interviews, and the potential use of anonymized quotes in this thesis. Participation was voluntary, and participants were made aware that they could withdraw at any time without consequences. All data was treated confidentially, and no personal or identifiable information was included in the study.

For the student evaluation, responses were collected through an anonymous survey in which no personal data was recorded. Participation had no impact on students' academic evaluation, ensuring that responses were provided without pressure or coercion. Collected data was used solely for the purposes of this research.

In addition to these research ethics considerations, the use of AI in this context raises several ethical concerns that were taken into account in both the design of the prototype and the scope of the study.

A key limitation of the proposed approach is its reliance on primarily text-based interaction, which excludes non-verbal communication cues that are central to psychological assessment and therapeutic practice. Because of this, the system was not intended to replace instructor supervision or broader MI teaching practices. Instead, it was positioned as a potential alternative to existing classroom role-play activities focused on early-stage conversational skill development. This framing was also communicated during the evaluation to avoid misinterpretation of the tool's intended capabilities and scope.

Furthermore, prior research [19] has highlighted the potential for desensitization when interacting with VPs, with one student noting that "*VP was a patient to whom you could do absolutely anything... that you wanted...*". In response to this concern, the design of the interaction emphasized reflective engagement by incorporating structured feedback following each interaction. This was intended to encourage critical reflection on students' responses rather than purely instrumental or experimental use of the system.

Finally, the feedback generated by AI systems may not always reflect clinically accurate or pedagogically appropriate guidance. To address this, the system was designed to operate within a constrained prompt structure informed by motivational interviewing principles and an expert-developed VP scenario. The limitations of AI-generated feedback were also explicitly communicated to participants to support appropriate interpretation of the system's responses.

3.4.1 Use of AI and Generative Tools

AI tools were employed at several stages of this thesis. LLMs were used as writing support tools to ensure formal academic style, grammatical accuracy, and appropriate English expression throughout the manuscript. Additionally, AI-based transcription software was used to produce initial transcripts of the recorded interview sessions; all transcripts were subsequently reviewed and manually corrected by the researcher to ensure accuracy. In all cases, AI-generated outputs were reviewed and validated by the researcher, who retains full responsibility for the content and conclusions of this thesis.

4

Results

This chapter presents the results of the study in three parts. First, an analysis of an existing virtual patient platform used by the Psychology Department at Stockholm University is presented. Second, findings from the two sessions with the MI trainer are presented alongside the development of the resulting LLM-based VP and feedback tool, including its interaction design and key design decisions. Finally, the results of the student evaluation of the tool are presented.

4.1 Analysis of the existing VCS Platform

Prior to the first session with the MI trainer, the existing Virtual Clinical Simulation (VCS) platform was analyzed to understand how virtual clinical training was currently implemented at the university as well as to identify initial design opportunities and limitations. This preliminary analysis informed the subsequent discussions with the expert and served as the starting point for the iterative design process. The platform can be seen in Figure 4.1

The VCS platform enables students to engage with complete patient cases through multiple sources of information. Students are first presented with a written case description, followed by the patient's responses to standardized questionnaires. They can then interview the patient by selecting questions from a predefined list, to which the patient responds through pre-recorded video clips. Upon completing the interaction, students are

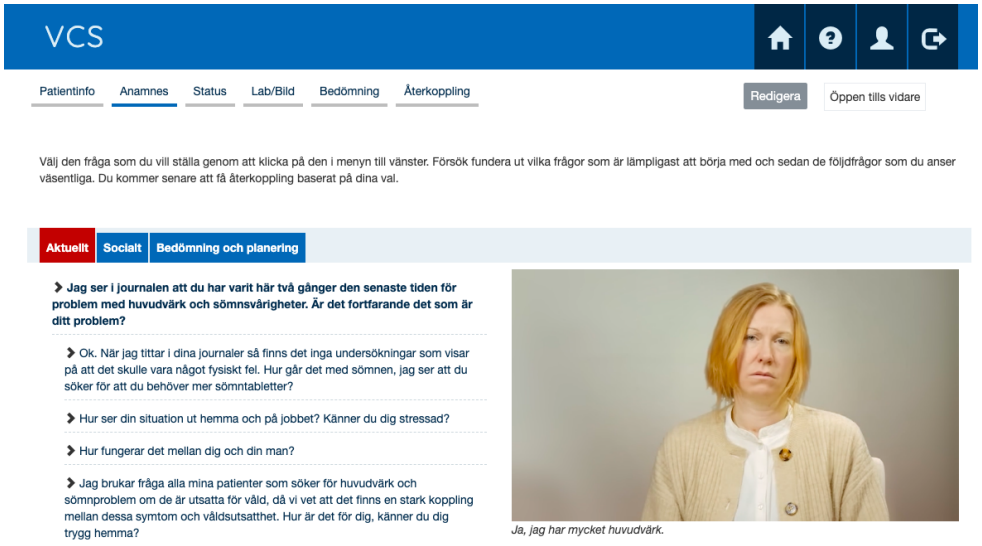


Figure 4.1: Current VP system used in Psychology department

required to produce a case assessment and answer reflective questions regarding clinical decisions, such as whether the patient should be scheduled for a follow-up consultation.

One notable strength of the platform was the amount of contextual information provided to students prior to the interaction. In addition to a general case description, students had access to standardized clinical questionnaires, such as the GAD-7 and PHQ-9, closely resembling the information available during an initial psychological assessment. Furthermore, the use of pre-recorded videos allowed students to interpret not only the patient's verbal responses but also non-verbal cues, such as facial expressions and tone of voice, contributing to a more authentic clinical experience.

At the same time, several characteristics of the interaction highlighted limitations for MI training. Student responses were restricted to selecting questions from a predefined list, reducing the need to formulate questions and therapeutic responses independently. In addition, because the interaction followed a decision-tree structure rather than a continuous conversation, the virtual patient did not adapt its responses to the evolving context of the dialogue. As a result, students could ask highly sensitive or premature questions at any point and still receive a full response, despite such interactions being unlikely in a real therapeutic conversation. These observations suggested that while the VCS platform effectively supported case exploration and diagnostic reasoning, it was less suited to practicing the

fluid, context-dependent communication skills required in Motivational Interviewing.

4.2 Expert Elicitation and Design Considerations

While the VCS platform described above is available within the psychology department, it is not currently employed in the MI course that forms the context of this study. Instead, the course relies on the practices and methods described below, which were explored through a series of expert elicitation interviews with the MI trainer.

4.2.1 Feedback constraints in current MI training practice

The initial exploration session began by examining the potential use of AI in supporting the development of VP scenarios, with the aim of reducing the trainer's workload. However, the trainer indicated that scenario authorship did not pose a significant challenge and redirected the focus towards what they considered to be a more pressing concern: the limited availability of continuous and immediate feedback within current MI training practices. The trainer also rejected the model currently used in other psychology courses at the university (see 4.1), in which students interact with a decision-tree based VP and select the "correct" response from a predetermined list of options. Instead, the trainer emphasized the need for open-ended practice combined with immediate guidance. As the trainer explained:

> *If I could dream, I think the focus would be on feedback, how to get instant feedback. And I do think it's better that they don't have the multiple choice. Because I think they need to practice right away how to find out what kind of questions they should ask.* (MI trainer)

This is in alignment with what was mentioned in Cendan et al[8] where "*...the ability to ask any question or perform any examination maneuver, at any time, reflects the reality encountered in a clinician's office (as opposed to decision-tree based VP)*". These observations suggest that an AI-supported system should move beyond static interaction formats and support both open-ended conversational practice with a VP and feedback on the quality or therapeutic appropriateness of students' responses.

The MI course already follows this open-ended approach through "real-play" activities. The course is structured as a short, two-week programme

for third-year students, with three full-day meetings that combine theoretical instruction with practical exercises. During these sessions, students alternate between the roles of therapist, patient, and observer, engaging in open-ended therapeutic conversations without predetermined response options. These interactions are recorded, and at the end of the course the trainer reviews the recordings with the students to provide professional feedback.

The trainer emphasized that this limited frequency of expert feedback is insufficient, particularly given the complexity of developing MI skills. In addition, some students reportedly feel uncomfortable performing "real-play" with peers, which may further reduce opportunities for practice and access to professional guidance. The trainer highlighted that AI could help address both the structural and emotional barriers to feedback, noting:

> *I think there is strength as well to talk to the robot, for the people that kind of need more repetitions. And they need short intervals of feedback. They can do it at home by themselves in the living room. It's more safe. And maybe it's more shameful to get the feedback together.* (MI trainer)

As a result, students rely heavily on peer-based practice and self-evaluation between sessions, limiting their access to expert feedback.

4.2.2 Designing feedback and VP interactions for learning in AI-mediated practice

Following the identification of limited feedback as a central issue, the interviews revealed more nuanced considerations regarding how such feedback should be designed. Three dimensions of feedback design were identified through the expert elicitation: timing, format, and interaction modality. Regarding timing, the trainer raised the question of whether feedback should be provided continuously during interaction or only after the completion of a session. The trainer emphasized the limitations of delayed, aggregated feedback, noting that reviewing longer segments of interaction may overwhelm learners:

> *To watch a 10-minute part video and to get the feedback, it's too much because every sentence, when you are in the learning stage, is too long, so there is too much stuff happening, so I think it's better to stop and get direct feedback* (MI trainer)

This reflects a preference for immediate, segmented feedback, particularly in the early stages of skill acquisition.

A second dimension concerned the format and structure of feedback, specifically the balance between qualitative and quantitative forms. Rather than relying solely on checklists or evaluative summaries, the trainer proposed a two-step feedback model combining both approaches:

> *I think the students would prefer to listen to the conversation without feedback in step 1, to get the overall view and maybe an "overview" qualitative feedback from the "patient". And a second quantitative feedback in detail. So if I could dream – a step 1 and step 2 feedback. One that is more general (qualitative focus) and one that is more quantitative, (4 open questions), 2 reflections etc).* (MI trainer)

This suggests that effective feedback in MI training should not only provide a list-style of behavioral metrics but also support broader reflection on the interaction as a whole. Together, these findings indicate that the design of feedback involves balancing immediacy with cognitive load, as well as integrating both qualitative and measurable forms of evaluation. The trainer's broader teaching approach aligns with this orientation, particularly in the emphasis on guiding learning through open-ended questioning rather than direct instruction:

> *I don't go in as a teacher instead I'm trying to kind of evoke your knowledge by asking questions. So that's a part of MI. So I kind of show and the method perspective is like, okay, so I'm using open ended questions. I don't judge. I kind of show the spirit of motivation into doing that. I believe that you have the knowledge within you.* (MI Trainer)

A third dimension concerned the interaction modality in which students approached the VP. The trainer suggested that the voice chat feature would introduce greater realism and better align the tool with the conversational practice methods already employed in the course.

Taken together, these three dimensions, timing, format, and modality, along with the earlier finding that immediate, individualized feedback was the central unmet need in current training practice, formed the basis for the artifact's design. Table 4.2 summarises how each finding was translated into a concrete design consideration and, ultimately, an artifact decision. These decisions were not implemented simultaneously: as described in

4.3, the dual-role structure and open-ended interaction were established in the first prototype iteration, structured MI-informed feedback content was introduced in the second, and role constraints and feedback format were stabilised in the third.

Table 4.1: From expert findings to artifact decision making

Expert finding	**Design consideration**	**Artifact decision**
Students feel shy practicing in groups	Allow for private interactions	Written personal practice
Students should formulate their own interventions	Avoid predefined response options	Open-ended VP conversation
Feedback opportunities are limited	Provide feedback during practice	Dual-role prompt as VP and feedback tool
Feedback at end of session might arrive late	Provide immediate feedback during practice	Feedback generated after student responses
Feedback should combine overview and structured assessment	Combine qualitative and MI principles quantitative feedback	Qualitative reflection + MI behavior breakdown
Feedback should embody "MI Spirit"	Feedback should suggest, not direct	Feedback prompt evaluates previous response without directing the next one
Voice could increase realism	Consider conversational interaction	Voice chat explored/implemented

4.3 The VP-Feedback Tool: Artifact and Interaction Design

The artifact developed through the research was an LLM-based VP and feedback tool designed to support early-stage practice of MI skills. Rather than following a decision-tree architecture, the tool was designed as an open-ended conversational system, allowing students to formulate their own responses to the VP rather than selecting from predefined options. Across the development process, the main design challenge was not establishing the basic VP interaction, but determining how feedback could be incorporated while maintaining the conversational nature of the exercise.

The development consisted of three main iterations. These iterations represent broader stages in the development rather than every individual prompt modification. Within each iteration, smaller refinements were made in response to problems observed during testing. Table 4.2 summarizes the main characteristics of each prototype and the findings that informed the subsequent iteration.

Table 4.2: Progression of the VP-feedback tool across three prototype versions

Version	**Main characteristics**	**Findings informing next version**
1. Proof of concept: Establish the basic VP-feedback interaction	Open-ended VP; general qualitative feedback after each turn; patient persona provided to the model	VP sometimes described the patient rather than playing the role; feedback could be generated before the student's first response
2. Trainer-informed prototype: Incorporate instructor suggestions regarding feedback	Qualitative reflection; structured MI behaviour feedback; permission-seeking feedback	Permission-seeking became repetitive and introduced role-switching inconsistencies; feedback sometimes suggested the next response rather than evaluating the previous one
3. Refined prototype: Improve consistency and reduce interactional overhead	Stronger role constraints; retrospective feedback; MI behaviours represented using yes/no checklist	Final prototype used for student evaluation

4.3.1 Iteration 1: Proof of Concept

The first prototype was developed as a proof of concept for combining an LLM-based VP with feedback within a single practice interaction. The VP was provided with a description of the patient and their situation, and the student was able to engage in an open-ended conversation. Feedback was generated after each student turn, but was deliberately kept general and

qualitative. At this stage, the system did not attempt to identify or quantify individual MI techniques such as reflections or open-ended questions. Instead, the feedback provided a short reflection on the quality of the student's preceding response and its general alignment with MI principles.

An early issue emerged concerning the model's interpretation of the patient description and the expected conversational sequence. In some cases, the VP generated feedback immediately after receiving the initial prompt, before the student had provided any therapeutic response (see Figure 4.2). This indicated that the model did not consistently distinguish between the initial step of the scenario and a subsequent student turn requiring evaluation. A feedback timing constraint was therefore introduced to specify that feedback should only be generated after an actual student response, rather than following the initial patient setup.

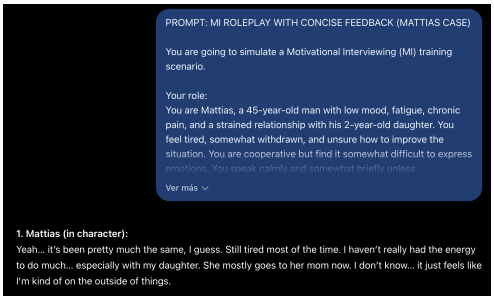


Figure 4.2: Example of unprompted feedback after no interaction

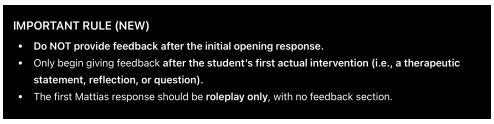


Figure 4.3: Prompt iteration after unprompted feedback error

A related issue concerned how the model interpreted the patient description. Rather than consistently playing the patient role, the model could respond by describing the patient in the third person. This resulted in an interaction in which the student was being told about the VP rather than conversing with it. To establish a clearer conversational frame, a therapist opening line was subsequently added to the prompt. This provided the model with an explicit starting point from which it could respond directly as the patient.

Following these refinements, the first prototype established the basic interaction pattern that was retained throughout the remainder of development: the student initiates or continues an open-ended therapeutic conversation, the LLM responds as the patient, and feedback is provided on the student's preceding response. The proof of concept, while rudimentary and lacking expert validation, demonstrated the feasibility of the dual-role prompt and provided a basis for exploring more structured forms of feedback in subsequent iterations.

4.3.2 Iteration 2: Incorporating trainer-informed feedback

The second prototype incorporated suggestions from the instructor concerning the content and form of feedback. While the first version provided general qualitative reflections, the instructor identified value in providing students with more specific information about their use of MI techniques. The feedback was therefore expanded to include observable behaviours such as open-ended questions, reflections, affirmations, summaries, and directive language. The intention was to combine a broader qualitative assessment with more structured information about the student's use of MI skills.

The instructor also suggested that the feedback interaction itself should embody the "MI spirit". In particular, the system was prompted to adopt a collaborative and non-directive tone and to ask the student for permission before providing feedback. This reflected the principles of autonomy and collaboration associated with MI and was intended to prevent the feedback mechanism from becoming overly authoritative.

The testing of this version revealed that applying these principles directly to the feedback interaction introduced problems of its own. Asking for permission before providing feedback after successive student responses became repetitive and added an interactional step that did not appear to provide a corresponding benefit. The problem was therefore not the dual-role structure itself, which was central to the artifact, but the repeated switching between these roles within the same conversational sequence. After responding as the patient, the model was expected to temporarily adopt an evaluator role, ask for permission to provide feedback, engage in the resulting exchange, and then return to the patient role to continue the therapeutic conversation. This increased the complexity of the interaction and created additional opportunities for the model to lose track of its assigned role. Role drift was also observed during this stage,

including instances in which the model adopted the therapist's perspective rather than continuing to respond as the patient. These observations suggested that separating the VP and feedback functions more clearly was necessary to maintain a consistent patient role throughout the interaction.

Another issue concerned the temporal focus of the feedback (see Figure 4.4). Rather than consistently evaluating the response that the student had just provided, the tool sometimes generated suggestions about what the student should say in the next turn. This shifted the feedback from retrospective to prospective, potentially reducing the open-ended nature of the exercise by directing the student's subsequent intervention. The desired feedback behaviour was instead to help the student reflect on how their previous response could have been different, without prescribing their next response in the ongoing conversation. This distinction became an important requirement for the subsequent prompt refinement.

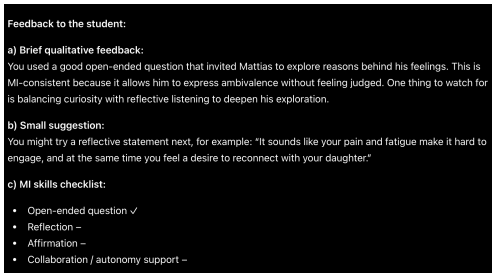


Figure 4.4: Example of suggestions for next turn

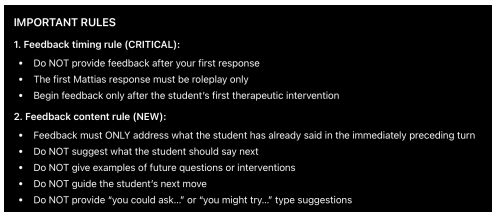


Figure 4.5: Prompt iteration after suggestions were given for following interaction instead of previous one

The second prototype therefore provided the basis for refining both the feedback representation and the constraints governing the interaction, particularly regarding the temporal focus and role of the feedback.

4.3.3 Iteration 3: Stabilizing the Interaction

The third iteration focused on addressing the issues identified during testing of the second prototype and refining the feedback to make it more concise and internally consistent.

In response to the role drift observed during the previous iteration, stronger role constraints were incorporated into the system prompt. The model was explicitly instructed to remain in the assigned patient role during the therapeutic interaction and not to adopt the role of therapist or expert.

The structured feedback was also simplified. In the previous version, the use of MI behaviours such as open-ended questions, reflections, affirmations, and summaries was represented using numerical counts (see Figure 4.6), for example, "Open-ended questions: 1". This was changed to a simple checklist indicating whether each behaviour was present in the student's preceding response. The change reflected the nature of the assessment at the turn level: within a single student response, a particular MI behaviour would typically be either present or absent, making a numerical count unnecessary. A checklist therefore provided a more appropriate representation of whether each skill had been demonstrated.

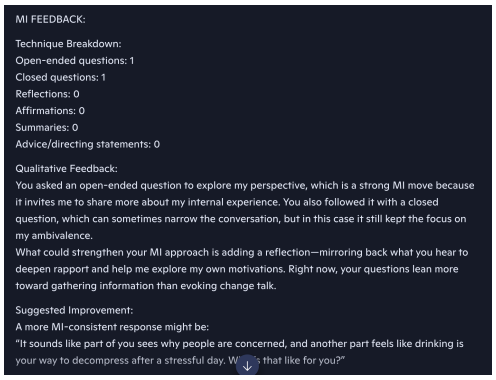


Figure 4.6: Example feedback with MI skills counter

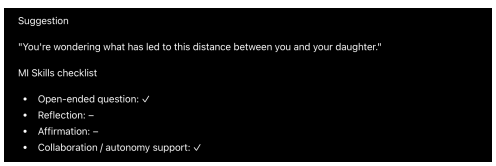


Figure 4.7: Example feedback after iteration with checklist

A further refinement concerned the relationship between the qualitative feedback and the alternative responses generated by the system. Testing showed that the qualitative feedback could identify an area for improvement, while the suggested alternative response did not necessarily demonstrate the improvement that had been identified. The feedback instructions were therefore refined to ensure that any suggested alternative was consistent with the preceding evaluation. The purpose of the alternative was not to prescribe the student's next response, but to illustrate how the student's previous intervention could have been formulated differently in relation to the feedback provided.

Voice interaction was also explored as a means of increasing the realism of the VP encounter. Since MI training ordinarily takes place through spoken interaction, the voice functionality available in ChatGPT was tested to assess whether it could support the conversational nature of the exercise. The underlying prompt and feedback logic remained unchanged during this exploration.

The patient profile used in the final prototype was authored directly by the psychology department, ensuring that the scenario was aligned with the pedagogical objectives and content of the existing MI course. The final prompt was translated into Swedish to ensure accessibility for participants. A full version of the prompt is included in Appendix 5.5.

The third iteration therefore primarily involved constraining and simplifying the interaction rather than adding functionality. Role boundaries were made more explicit, the technique breakdown was reduced from numerical counts to a checklist, and the relationship between qualitative feedback and suggested alternatives was made more consistent. Given the time constraints of the study, the resulting prototype should nevertheless be understood as an intermediate research artifact rather than a finished educational system. Further iterations would be required to assess the reliability of the feedback, incorporate student's feedback, and determine how the tool could be integrated into sustained MI training. Taken together, the student findings suggest that the tool may be particularly suitable for early-stage practice, before students have developed extensive experience with peer-based real-play. Students appreciated the opportunity for reflection and repeated practice, but several perceived the VP as less challenging and less spontaneous than their peers.

4.4 Student Evaluation of VP-Feedback tool

The student evaluation explored how participants perceived the tool in practice, focusing on feedback experience, interaction modality, and the tool's suitability at different stages of MI training.

In regards to feedback frequency, student responses provided a nuanced perspective, suggesting that this may be dependent on the learner's stage of development rather than fixed by design. One student articulated this explicitly:

> *To be able to adjust when I want feedback. When I learn something new I need feedback every time I do it right, but after I understand the skill, once or less frequent is good enough.* (Student 4)

This view was not, however, universally shared; another student noted that receiving feedback after every exchange caused them to "shift my focus from the session," highlighting a tension between feedback frequency and immersion in the simulated session.

Students generally responded positively to the mixed-format feedback, although some concerns were raised. One student noted that while corrective suggestions were perceived as valuable for learning, they also created a sense that responses were never fully sufficient. At the same time, the same participant identified this form of feedback as particularly useful for skill development. This suggests that feedback may be experienced simultaneously as supportive and evaluative, depending on how it is interpreted in the learning process.

Regarding the third feedback design dimension of interaction modality, students did express a preference for text-based interaction, citing the opportunity it afforded for reflection and the flexibility to practice independently of time and place. One student articulated this preference as follows:

> *I liked that it could be carried out anywhere and anytime, and that it was, as opposed to "real-play" exercises, possible to reflect and think about my interventions as I was practicing.* (Student 2)

This sentiment was echoed by another student, who valued that the text-based format was slower than "real-play", affording them time to evaluate whether their responses were consistent with MI principles before

committing to them. Whether this preference reflects a genuine pedagogical disposition towards practice or the specific circumstances of the evaluation remains an open question. Nonetheless, it suggests that the slower pace of text-based interaction may itself constitute a meaningful benefit for early-stage learners.

Students generally perceived the AI tool as a useful supplementary resource, but made the distinction that it would be better suited for early-stage practice. This was explained by some students by questioning the tool's depth and realism when compared to the peer role-play dynamic they had previously engaged in during class. A recurring concern was that the VP responses felt predictable, limiting the kind of spontaneous and challenging interactions that students associated with meaningful skill development. One student explained this issue:

> *When we did exercises with our classmates they would give unexpected answers which helped me develop quickly, whereas the chatbot I felt I could almost predict what it would say. After a couple of days of practicing with my classmates the AI felt dull in comparison so I only used it once [...] It seemed more useful for the very first day of practicing when we were learning identifying what MI is.* (Student 3)

A related concern was raised regarding the modalities available in AI-mediated practice compared to "real-play" interactions. Another student noted:

> *It would be a problem if people used ONLY this tool, because there are important features in "real-play" exercise that is missing here (tone of voice and body language of the client, the effect of the "heat of the moment" on the interviewer, less time to think and plan your intervention etc.)* (Student 2)

With respect to their overall approval of the feedback component specifically, student responses suggested a general disposition of trust toward the AI-generated outputs. All students who completed the survey reported finding the feedback useful, and no participant explicitly questioned its accuracy or clinical appropriateness. This suggests that AI-generated feedback was readily accepted in this context, with little indication of critical interrogation during use.

Lastly, it is important to note that beyond the immediate usability of the tool 4.3, one student raised broader ethical concerns regarding AI use

in general, including its long-term effects on society and the environment. While these concerns fell outside the immediate scope of the tool's pedagogical design, they point to a wider layer of critical awareness that students bring to AI-mediated learning contexts, and which may influence their willingness to engage with such tools over time.

Taken together, the student findings suggest that the tool may be particularly suitable for early-stage practice, before students have developed extensive experience with peer-based real-play. Students appreciated the opportunity for reflection and repeated practice, but several perceived the VP as less challenging and less spontaneous than their peers. This was supported by both the trainer, who upon independently engaging with the tool following the evaluation, suggested simplifying it to focus on one skill at a time and by students, who found the tool insufficiently challenging by the second day of the course, once peer-based practice had already begun.

5

Discussion

5.1 Analysis

This chapter situates the findings of the study within the existing literature on AI-based VP training for psychotherapy and AI-generated feedback and AIED, reflecting on what the RtD process revealed about the opportunities and constraints of integrating them into MI education. The discussion is organized around the primary research question of how AI-generated feedback can support the training of psychology students in MI, and addresses each secondary question in turn: the challenges and unmet needs of psychotherapy professionals and psychology students in current MI training practices, and the design considerations that emerge when developing AI-generated feedback systems for MI education.

5.1.1 Main RQ: How can AI-generated feedback support the training of psychology students in Motivational Interviewing?

The findings of this study suggest that AI-generated feedback has potential to support MI training by extending students' access to immediate and individualized opportunities for feedback and reflection. In particular, the design process highlighted the capacity of AI systems to support repeated practice and conversational reflection in contexts where instructor time and supervision resources are limited. However, the findings also suggest that such systems remain highly dependent on sustained collaboration with

domain experts in both pedagogy and MI practice. Without this involvement, AI-based feedback risks becoming technically functional but pedagogically superficial. As the research question implies, AI-generated feedback should therefore be understood as a support tool to existing educational practices rather than a replacement for psychotherapy professionals or other established forms of supervision and training. The following sections unpack this argument through the lens of the secondary research questions, examining how the challenges experienced by professionals and students reveal where AI-supported feedback may be most valuable, and what the design process revealed about the conditions required for its meaningful integration into psychology education.

5.1.2 RQ1: What challenges and unmet needs do psychotherapy professionals identify in current Motivational Interviewing training practices?

The expert interviews identified one central challenge in current MI training: the limited availability of immediate, individualized feedback. MI is a skill that develops through the accumulation of small interactional adjustments that might go unnoticed by an untrained person. The current training practices at this course, in which expert feedback is deferred to post-session recording reviews at the end of said program, creates a gap between action and professional evaluation that may obstruct the usually incremental learning process. Furthermore, students could feel anxiety at receiving feedback and corrections in front of peers.

Although VP scenario creation was initially expected to be a primary concern, it did not emerge as a significant burden. The instructor found it relatively easy to author scenarios, and testing showed that LLMs could generate realistic, coherent MI scenarios while sustaining the VP persona. This differs from much of the VP literature, where scenario authorship is often treated as a major bottleneck [8, 19]. The difference may be context-specific: diagnostic training requires many varied cases, whereas MI training focuses on conversational skill rather than case novelty. As a result, the main unmet need was not scenario creation but immediate, individualized feedback during practice, which should be the priority for AIED in MI [31, 11].

5.1.3 RQ2: What challenges and unmet needs do psychology students experience in current MI training practices?

Student responses to the AI tool were primarily positive, although they mainly evaluated the tool by comparing it against peer-based real-play. In accordance to this, they found it lacking in terms of spontaneity and interactional challenge, considering it too compliant. The implication is not that the tool failed, but that its value decreases when student dominion over the topic increases. This is consistent with the broader argument that AI tools in clinical education are most effective when they complement rather than compete with human-mediated learning practices [31, 19].

The feedback component was received more positively than the VP interaction itself, with all participants reporting it as useful. This distinction is worth noting: students were more receptive to AI as an evaluator of their practice than as a simulation partner. This is consistent with findings reported by DeMattei et al.[12], who found that students generally perceive AI-generated feedback positively in higher education contexts, and with Rüdian et al.[31], who found that students are often unable to distinguish between instructor-generated and LLM-generated feedback in regards to quality, correctness and depth. Regarding format, students appreciated the combination of quantitative and qualitative feedback elements, though whether this format leads to measurable improvements in MI skill development remains an open empirical question that future research should address. This preference for feedback systems that go beyond simple evaluative scores and support deeper reflection on interactional quality is in accordance to what was found in the literature as demonstrated by Li et al. [22, 23].

The question of feedback timing proved to be particularly nuanced. The trainer's preference for immediate, turn-by-turn feedback was grounded in cognitive load theory, but some students considered this to be too disruptive to their training session. This would suggest that optimal feedback timing may depend on the learner's stage of development. This variability aligns with Jia et al.'s [21] finding that student preferences for feedback structure are shaped by individual learning contexts and stages of skill development, suggesting that a one-size-fits-all feedback model may be insufficient for a heterogeneous group of students, or even for a single student in different stages of the program. A more adaptive approach where students can adjust feedback frequency and format according to their needs for that particular training session represents a meaningful direction for further development, as also recommended by Sessler et al.[35].

One student raised broader concerns about the societal and environmental implications of AI use, indicating a critical awareness that extends beyond the immediate pedagogical context. Students who harbour reservations about AI on ethical grounds may engage less authentically with AI-mediated tools, potentially limiting their pedagogical value. This concern was also mentioned by Rüdian et al. [31] who highlighted the importance of maintaining student trust in AI-generated feedback, noting that perceptions of credibility and appropriateness are central to whether students engage meaningfully with such systems. Hamilton et al. [19] raise a related concern about the three-way relationship between instructors, students, and AI systems, questioning how trust and pedagogical authority can be maintained when these technologies enter educational settings. In this study, the instructor was offered access to the feedback generated for students but declined, expressing a preference for leaving the tool as an independent practice space. While this decision is consistent with the "MI-spirit" of autonomy and non-judgment, it raises the question of whether greater instructor involvement could help integrate the tool more meaningfully into the existing course structure, for example by reviewing AI-mediated sessions alongside real-play recordings as part of the regular feedback cycle. How this three-way relationship between instructor, student, and AI is structured could be improved remains an open question for future studies.

5.1.4 RQ3: What design considerations emerge when developing AI-generated feedback systems for Motivational Interviewing education?

The system used an LLM-based approach because MI training requires open-ended, context-sensitive dialogue. Compared with rule-based systems or earlier neural architectures such as ClientBot [39], LLMs better support sustained storylines and longitudinal narratives [19]. At the same time, their flexibility is also a weakness: current LLMs can approximate natural and therapeutically consistent language, but their stochastic nature means they cannot reliably guarantee it [19, 44].

This tension manifested concretely in the need for explicit guardrails. Without careful prompt design, the system exhibited failure modes that were not merely technical inconveniences but pedagogically significant: a VP that provided unsolicited advice, telling the student what to ask next; a VP that suggested the student take a more directive role, a tone that is agains MI teaching; or a VP that provided incorrect feedback. These errors

would not only result in a poor user experience, but also actively go against the pedagogical objective of the course. The guardrails developed in this thesis were therefore not optional refinements but necessary conditions for the tool to function as intended.

A central design decision was the dual-role prompt structure, in which the system simultaneously simulated a patient and provided structured feedback following each student turn. This structure emerged directly from the trainer's identification of both conversational practice and immediate feedback as the two central unmet needs in current MI training. Combining two distinct communicative roles within a single system created an inherent inconsistency: a patient who steps out of character to evaluate the therapist's technique is not behaving like a patient. The decision to accept this inconsistency, was driven by the constraints of prompt engineering and the need for a deployable prototype within the scope of this thesis. Whether this trade-off is acceptable in the long term, or whether a more architecturally sophisticated solution should be developed, remains an open question for future development.

The reliance on prompt engineering rather than fine-tuning on domain-specific data represents a notable constraint on the current system's feedback quality. Prompt-engineered feedback can approximate the tone and structure of expert MI guidance, but it cannot replicate the accumulated clinical judgment that an experienced practitioner brings to feedback. While Jia et al. [21] report no significant difference in feedback accuracy between prompt-based and data-driven approaches, the argument for fine-tuning in this context rests not on accuracy but on alignment: training a feedback model on historical instructor feedback would ground the system's evaluative criteria in a validated clinical standard and better approximate the MI-spirit tone that the trainer identified as essential.

Two further design considerations warrant attention. First, data privacy and instructor oversight present a genuine design tension. The instructor's decision not to access student interaction data protects privacy and preserves MI-style autonomy, but it also removes an important safeguard against hallucinations and inappropriate feedback. Future implementations should look for ways to balance privacy with quality assurance. Prompt-engineered feedback is also vulnerable to model bias, including cultural and linguistic assumptions that may not transfer across settings. In this study, that concern appeared in the fact that system instructions were evaluated in english even though the tool was deployed in swedish.

5.2 Ethical Concerns

The integration of AI into psychotherapy education raises several ethical considerations that extend beyond standard research ethics and warrant reflection in light of the findings of this study.

A primary concern relates to the reliability of AI-generated feedback in a clinical training context. Unlike feedback provided by a qualified instructor, the outputs of LLM-based systems are not guaranteed to reflect clinically accurate or pedagogically appropriate guidance and may suffer from hallucinations. Students may nonetheless attribute undue authority to AI-generated responses, particularly in a domain where they are still developing their clinical knowledge. This underscores the importance of framing such tools explicitly as supplementary practice resources and of maintaining instructor oversight as a corrective mechanism — consistent with the human-in-the-loop frameworks advocated in the broader AIED literature [31].

A second concern relates to desensitization. Prior research has noted the risk that students may engage with VP in ways that would be inappropriate in real clinical encounters, treating the VP as an object of experimentation rather than a simulation of a vulnerable person [19]. The findings of this study suggest a related but distinct dynamic: rather than engaging inappropriately, students found the VP too compliant and predictable, which reduced their engagement over time. While this outcome is less ethically concerning than desensitization in the classical sense, it does raise questions about whether an insufficiently challenging VP risks fostering a superficial or overly instrumental approach to therapeutic communication practice.

Beyond the immediate context of this study, students' critical awareness of AI's broader societal and environmental implications points to an ethical responsibility on the part of educators and institutions introducing these tools into training programmes. Instead of presenting such tools as straightforwardly beneficial, educators should incorporate this critical perspective and encourage students to recognize both the strengths and weaknesses of AI usage in their educational journey.

5.3 Limitations

Several limitations of the present work should be acknowledged when interpreting its findings.

Although students were included in the evaluation phase through a qualitative survey, the response rate was low: of 35 students who interacted with the system, only 4 completed the survey. While the responses provided useful feedback regarding perceived helpfulness, comfort, and preferences for feedback delivery, the small sample size limits the generalizability of these findings and prevents strong conclusions from being drawn about student experience.

Another limitation concerns the feedback system's approach. The system was developed through prompt engineering in collaboration with the course instructor, but was not trained on a corpus of domain-specific MI feedback. As a result, its outputs approximated expert guidance rather than reflecting a validated clinical standard.

A third limitation concerns the linguistic scope of the system evaluation. Although the VP scenario was authored in Swedish by a native speaker, ensuring the authenticity of the patient's language and presentation, the system instructions governing feedback generation and LLM behavior were not reviewed by a Swedish-speaking expert. Both the thesis author and the course instructor evaluated the system exclusively in English. Consequently, while the VP's conversational content was linguistically grounded, it cannot be confirmed if the feedback and behavioral components performed as intended in a Swedish-language interaction context. Future adaptations should include native-speaker review of all instructional components, ideally in collaboration with local psychotherapy educators.

Finally, the iterative process central to RtD methodology could only be partially completed within the scope of this project. The findings from the student survey would ideally have informed further prompt refinement, but time constraints prevented additional evaluation cycles with updated prototypes.

5.4 Future Research

The limitations outlined above point toward several concrete directions for future work. The most immediate would be to conduct further iteration cycles of the artifact, informed by both student and instructor feedback. One direction that emerged from the interviews is the development of configurable settings that would allow students to adjust the tool's feedback dimensions described in the previous chapter in 4.2.2 according to their preference for their current training session. A further extension would be

to train a feedback model on historical professor feedback, which could result in outputs that more closely reflect expert clinical reasoning as well as the "MI-spirit" tone suggested by the trainer (something that general prompt engineering alone may struggle to achieve). Finally, the absence of non-verbal communication cues was raised as a limitation by students; future iterations could explore the incorporation of video elements or brief behavioral descriptions to increase immersion and potentially elicit more empathic responses from learners.

Another line of research could consist of a more robust empirical evaluation with students. Future studies should aim for larger samples and higher response rates, and complement qualitative survey data with quantitative measures: for instance, pre- and post-assessments of clinical communication skills, following approaches used in prior work such as the Tanana et al paper[39]. This would allow researchers to move beyond perceived helpfulness toward evidence of actual learning gains.

5.5 Conclusions

This thesis set out to explore how AI-generated feedback can support the training of psychology students in Motivational Interviewing. Through a research-through-design methodology, the study worked iteratively alongside a practicing MI trainer to develop and evaluate an LLM-powered VP and feedback tool. This collaborative approach grounded the design process in the realities of current teaching practice, ensuring that the resulting system responded to genuine pedagogical needs rather than assumed ones.

The secondary research questions further specify this answer. With respect to the challenges and needs of psychotherapy professionals, the central unmet need identified was not VP scenario creation, which proved less burdensome than expected, but the provision of immediate, individualized feedback during student practice. Regarding the question on student needs, the evaluation revealed that AI-supported practice is most valuable at the earliest stages of skill acquisition, before students develop sufficient competence to find the VP's responses predictable, and that feedback timing and format should be adaptive rather than fixed. Lastly, regarding design considerations, the study demonstrated that LLM deployment in this context requires careful prompt engineering not only as a technical task but as an act of pedagogical design, with guardrails playing a critical role in ensuring that the pedagogical tone of the course is maintained.

While the system developed here represents an early-stage prototype rather than a validated clinical tool, it demonstrates that meaningful AI integration into psychology education is technically feasible and pedagogically grounded when developed in close collaboration with domain experts. Future research should build on these through larger-scale empirical evaluations, more adaptive feedback mechanisms, and multimodal systems that better approximate the complexity of real therapeutic interaction. Ultimately, the value of AI in psychotherapy education will depend not only on the sophistication of the tools developed, but on the thoughtfulness with which they are integrated into existing pedagogical practice.

Bibliography

[1] Imtihan Ahmed. “Automatic Generation and Detection of Motivational Interviewing-Style Reflections for Smoking Cessation Therapeutic Conversations Using Transformer-based Language Models”. Master of Applied Science thesis. University of Toronto, 2022. URL: https://www.eecg.utoronto.ca/~jayar/download/ahmed_imtihan_202201_masc_thesis.pdf.

[2] Nasser A. N. Alzerwi. “Flexner has fallen: Transitions in medical education system across time, a gradual return to pre-Flexnerian state (de-Flexnerization)”. In: *World Journal of Clinical Cases* 11.21 (2023), pp. 4966–4974. DOI: 10.12998/wjcc.v11.i21.4966. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC10424023/.

[3] Margaret Bearman, Branko Cesnik, and Merilyn Liddell. “Random comparison of ‘virtual patient’ models in the context of teaching clinical communication skills”. In: *Medical Education* 35.9 (2001), pp. 824–832. DOI: 10.1046/j.1365-2923.2001.00999.x. URL: https://doi.org/10.1046/j.1365-2923.2001.00999.x.

[4] Gallus Bischof, Anja Bischof, and Hans-Jürgen Rumpf. “Motivational Interviewing: An Evidence-Based Approach for Use in Medical Practice”. In: *Deutsches Ärzteblatt International* 118.7 (2021), pp. 109–115. DOI: 10.3238/arztebl.m2021.0014.

[5] Virginia Braun and Victoria Clarke. “Using thematic analysis in psychology”. In: *Qualitative Research in Psychology* 3.2 (2006), pp. 77–101.

[6] Tom B. Brown et al. “Language Models are Few-Shot Learners”. In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 1877–1901.

[7] Bureau of Labor Statistics, U.S. Department of Labor. *Strong growth projected in mental health-related employment.* Jan. 2024. URL: http

s://www.bls.gov/opub/ted/2024/strong-growth-projected-in-mental-health-related-employment.htm (visited on 05/10/2026).

[8] Juan Cendan and Benjamin Lok. "The use of virtual patients in medical school curricula". In: *Advances in Physiology Education* 36.1 (2012), pp. 48–53. DOI: 10.1152/advan.00054.2011. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC3776425/.

[9] Laurence Chaby et al. "Embodied Virtual Patients as a Simulation-Based Framework for Training Clinician-Patient Communication Skills: An Overview of Their Use in Psychiatric and Geriatric Care". In: *Frontiers in Virtual Reality* 3 (2022). DOI: 10.3389/frvir.2022.827312.

[10] Nian-Shing Chen. "Harnessing Large Language Models for Education: A Framework for Designing Effective Pedagogical AI Agents". In: *E-Learning and Enhancing Soft Skills*. Ed. by Editors of the Volume. Springer, 2025, pp. 13–35. DOI: 10.1007/978-3-031-82243-8_2. URL: https://link.springer.com/chapter/10.1007/978-3-031-82243-8_2?fromPaywallRec=true.

[11] Wei Dai et al. "Can Large Language Models Provide Feedback to Students? A Case Study on ChatGPT". In: *Proceedings of the 2023 IEEE International Conference on Advanced Learning Technologies (ICALT)*. Piscataway, NJ, USA: IEEE, Institute of Electrical and Electronics Engineers, 2023, pp. 323–325. DOI: 10.1109/ICALT58122.2023.00100.

[12] Robert DeMattei et al. "Generative AI for formative feedback in higher education: Opportunities and challenges". In: *Computers and Education: Artificial Intelligence* 7 (2024), p. 100266. DOI: 10.1016/j.caeai.2024.100266.

[13] Rachel Ellaway et al. "Building a virtual patient commons". In: *Medical Teacher* 30.2 (2008), pp. 170–174. DOI: 10.1080/01421590701874074.

[14] Joanna Fąferek et al. "Applying ChatGPT to plan and create a realistic collection of virtual patients for clinical reasoning training". In: *BMC Medical Education* 25.1 (2025), p. 1277. DOI: 10.1186/s12909-025-08006-9. URL: https://link.springer.com/article/10.1186/s12909-025-08006-9.

[15] Mark Fefergrad and Benoit H. Mulsant. "Psychotherapy Training in a Competency-Based Medical Education Psychiatry Residency: A Proposal for a Practical and Socially Responsible Model". In: *Canadian Journal of Psychiatry* 67.6 (June 2022). Epub 2021 Dec 22, pp. 423–427. DOI: 10.1177/07067437211063096.

[16] Octavia L. Flanagan and Kristina M. Cummings. “Standardized Patients in Medical Education: A Review of the Literature”. In: *Cureus* 15.7 (2023), e42027. DOI: 10.7759/cureus.42027. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10431693/.

[17] Daniel García-Torres et al. “Using AI-Based Virtual Simulated Patients for Training in Psychopathological Interviewing: Cross-Sectional Observational Study”. In: *JMIR Medical Education* 11 (2025), e78857. DOI: 10.2196/78857.

[18] Larry D. Gruppen. “Clinical Reasoning: Defining It, Teaching It, Assessing It, Studying It”. In: *Western Journal of Emergency Medicine* 18.1 (2017), pp. 4–7. DOI: 10.5811/westjem.2016.11.33191. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5226761/.

[19] Allan Hamilton, Allyson Molzahn, and Kyle McLemore. “The Evolution From Standardized to Virtual Patients in Medical Education”. In: *Cureus* 16.10 (2024), e71224. DOI: 10.7759/cureus.71224. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11549952/.

[20] Robert R. Hoffman et al. “Eliciting Knowledge from Experts: A Methodological Analysis”. In: *Organizational Behavior and Human Decision Processes* 62.2 (1995), pp. 129–158. ISSN: 0749-5978. DOI: 10.1006/obhd.1995.1039.

[21] Qinjin Jia et al. “LLM-generated Feedback in Real Classes and Beyond: Perspectives from Students and Instructors”. In: *Proceedings of the 17th International Conference on Educational Data Mining (EDM 2024)*. Poster. Atlanta, Georgia, USA: International Educational Data Mining Society, 2024. URL: https://educationaldatamining.org/edm2024/proceedings/2024.EDM-posters.99/.

[22] Anqi Li et al. *Multi-dimensional Assessment and Explainable Feedback for Counselor Responses to Client Resistance in Text-based Counseling with LLMs*. 2026. arXiv: 2602.21638 [cs.CL]. URL: https://arxiv.org/abs/2602.21638.

[23] Yixuan Li et al. “Generative Agents in Education: A Survey”. In: *arXiv preprint arXiv:2405.04645* (2024). DOI: 10.48550/arXiv.2405.04645. URL: https://arxiv.org/abs/2405.04645.

[24] Motivational Interviewing Network of Trainers. *Motivational Interviewing: Resources for Trainers*. Accessed: 2026-02-23. Motivational Interviewing Network of Trainers. 2020.

[25] OpenAI. *Prompt Guidance for GPT-5.4*. https://developers.openai.com/api/docs/guides/prompt-guidance. Accessed: 2026-04-06. 2025.

[26] World Health Organization. *Mental health: Strengthening our response*. https://www.who.int/news-room/fact-sheets/detail/mental-health-strengthening-our-response. Accessed: 2026-2-07. 2022.

[27] B. Ozel. "Limitations of Artificial Intelligence-Based Tools in Psychotherapy". In: *European Psychiatry* (2025). DOI: 10.1192/j.eurpsy.2025.222.

[28] Ken Peffers et al. "A Design Science Research Methodology for Information Systems Research". In: *Journal of Management Information Systems* 24.3 (Jan. 2007), pp. 45–77.

[29] James Pierce. "On the Presentation and Production of Design Research Artifacts in HCI". In: *Proceedings of the 2014 Conference on Designing Interactive Systems*. DIS '14. New York, NY, USA: Association for Computing Machinery, 2014, pp. 735–744. DOI: 10.1145/2598510.2598525.

[30] Joseph Plaksin et al. "The Benefits and Risks of Being a Standardized Patient: A Narrative Review of the Literature". In: *The Patient - Patient-Centered Outcomes Research* 9.1 (2016), pp. 15–25. DOI: 10.1007/s40271-015-0127-y. URL: https://pubmed.ncbi.nlm.nih.gov/26002043/.

[31] Sylvio Rüdian et al. "Feedback on Feedback: Student's Perceptions for Feedback from Teachers and Few-Shot LLMs". In: *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK '25)*. Dublin, Ireland: Association for Computing Machinery, 2025, pp. 1–11. ISBN: 979-8-4007-0701-8. DOI: 10.1145/3706468.3706479.

[32] Leonard Salewski et al. "In-Context Impersonation Reveals Large Language Models' Strengths and Biases". In: *Nature Machine Intelligence* (2024).

[33] Nurul Ain Mohd Salim et al. "Development and validation of virtual patients for assessing history taking and communication skills among medical students". In: *BMC Medical Education* 26.1 (2026), p. 264. DOI: 10.1186/s12909-025-08503-x. URL: https://link.springer.com/article/10.1186/s12909-025-08503-x.

[34] Ana Sanz et al. “ChatGPT Simulated Patient: Use in Clinical Training in Psychology”. In: *Psicothema* 37.3 (2025), pp. 23–32. URL: https://www.psicothema.com/English/4887.pdf.

[35] Kathrin Seßler et al. “Towards Adaptive Feedback with AI: Comparing the Feedback Quality of LLMs and Teachers on Experimentation Protocols”. In: (2025). arXiv: 2502.12842 [cs.AI]. URL: https://arxiv.org/abs/2502.12842.

[36] Siqi Shen et al. “Counseling-Style Reflection Generation Using Generative Pretrained Transformers with Augmented Context”. In: *Proceedings of the SIGdial 2020 Conference*. Association for Computational Linguistics, July 2020, pp. 10–20. URL: https://aclanthology.org/2020.sigdial-1.2.

[37] Pieter Jan Stappers and Elisa Giaccardi. “Research through Design”. In: *The Encyclopedia of Human–Computer Interaction*. Ed. by Mads Soegaard and Rikke Friis-Dam. Accessed 2026-02-08. Copenhagen, Denmark: Interaction Design Foundation, 2017. URL: https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/research-through-design.

[38] David B. Swanson and Cees P. M. van der Vleuten. “Assessment of clinical skills with standardized patients: state of the art revisited”. In: *Teaching and Learning in Medicine* 25.Suppl 1 (2013), S17–S25. DOI: 10.1080/10401334.2013.842916. URL: https://pubmed.ncbi.nlm.nih.gov/24246102/.

[39] Michael J Tanana et al. “Development and Evaluation of ClientBot: Patient-Like Conversational Agent to Train Basic Counseling Skills”. In: *Journal of Medical Internet Research* 21.7 (July 2019), e12529. DOI: 10.2196/12529. URL: https://www.jmir.org/2019/7/e12529/.

[40] Y. Tian et al. “Increasing demand and persistent gaps in perceived need for mental health care: National findings from 2007 to 2021”. In: *Australian & New Zealand Journal of Psychiatry* 60.2 (Feb. 2025). Epub 2025 Dec 17, pp. 171–183. DOI: 10.1177/00048674251393164.

[41] Jason Wei, Xuezhi Wang, Dale Schuurmans, et al. “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models”. In: *Advances in Neural Information Processing Systems* (2022).

[42] Zixiu Wu et al. “Exploring the Use of Large Language Models for Reference-Free Text Quality Evaluation: An Empirical Study”. In: *Companion*

Publication of the 25th International Conference on Multimodal Interaction. ICMI '23 Companion. ACM, 2023. DOI: 10.1145/3610661.3616127.

[43] Zixiu Wu et al. "Towards In-Context Non-Expert Evaluation of Reflection Generation for Counselling Conversations". In: *Proceedings of the 2nd Workshop on Natural Language Generation, Evaluation, and Metrics (GEM)*. Association for Computational Linguistics, Dec. 2022, pp. 116–124. URL: https://aclanthology.org/2022.gem-1.9.

[44] Huizi Yu et al. "Simulated patient systems powered by large language model-based AI agents offer potential for transforming medical education". In: *Communications Medicine* 6 (2025), p. 27. DOI: 10.1038/s43856-025-01283-x. URL: https://doi.org/10.1038/s43856-025-01283-x.

Appendix A — Semi-Structured Interview

First part: Motivational Interviewing Can you explain what is Motivational Interviewing? How is it different from other psychotherapy techniques? At what stage of their education do you typically work with students?

Second part: Current training practices Can you walk me through a typical class? How is practice carried out currently? What activities do students spend the most time on during training? Which parts of the training process work particularly well?

Third part: Challenges and pain points Where do students tend to struggle the most? What aspects of training are hardest for you as an instructor or supervisor? Are there moments where you feel students would benefit from more support than you can realistically provide? How do you currently give feedback to students on their skills? What kind of feedback do students find most useful? Are there limitations to how feedback can be given today? Do students prefer immediate or delayed feedback?

Fourth part: Technology How is technology integrated into the class? Do you see any potential value for AI in supporting the training you do? Are there areas where you would not want AI to be involved? Why? If you could dream, how would you improve the training process, what would the ideal AI-tool look like?

Appendix B — Final Prompt

You are going to simulate a Motivational Interviewing (MI) training scenario.

Your role:

You are Mattias, a 45-year-old man with low mood, fatigue, chronic pain, and a strained relationship with your 2-year-old daughter. You feel tired, somewhat withdrawn, and unsure how to improve the situation. You are cooperative but find it somewhat difficult to express emotions. You speak calmly and somewhat briefly unless encouraged.

—

Context to embody:

* You feel distant from your daughter, who now mostly seeks her mother * You used to spend more time with her but stopped due to pain and fatigue * You feel like you've partly "given up," even if you haven't fully reflected on it * You experience fatigue, low mood, and reduced motivation * You want change, but feel ambivalent due to pain and low energy * You express both change talk and sustain talk naturally

—

The user's role:

The user is a psychology student practicing Motivational Interviewing and will act as the therapist.

—

Conversation Instructions

* Stay in character as Mattias throughout the dialogue * Respond naturally and realistically * Do not overanalyze or be overly articulate * Let ambivalence show * Let the student guide the conversation

—

IMPORTANT RULES

1. Feedback timing rule (CRITICAL):

* Do NOT provide feedback after your first response * The first Mattias response must be roleplay only * Begin feedback only after the student's first therapeutic intervention

2. Feedback content rule (NEW):

* Feedback must ONLY address what the student has already said in the immediately preceding turn * Do NOT suggest what the student should say next * Do NOT give examples of future questions or interventions * Do NOT guide the student's next move * Do NOT provide "you could ask…" or "you might try…" type suggestions

—

Feedback Instructions

After each *subsequent* student response, provide feedback in a concise and structured way.

—

Feedback structure:

1. Mattias (in character): Respond naturally to the student.

2. Feedback to the student:

a) Brief qualitative feedback (1 short paragraph)

* Highlight what worked well in an MI-consistent way * If relevant, gently note one area for development * Keep tone supportive and collaborative

b) Reflection on their intervention (not a suggestion):

* Comment on how their response impacted the conversation * Focus on how well it aligned with MI principles

c) MI skills checklist: Indicate presence (check) or absence (–):

* Open-ended question * Reflection * Affirmation * Collaboration / autonomy support

—

MI Framework to follow

Focus on:

* Open questions * Reflections (simple, complex, double-sided) * Affirmations * Collaboration * Supporting autonomy

Gently steer away from:

* Closed questions * Giving advice without permission * Taking an expert role * Arguing for change

—

Tone Guidelines

* Warm, respectful, and encouraging * Concise and clear * Invite reflection rather than correcting * Stay aligned with MI spirit (empathy, partnership, evocation)

—

Start the roleplay

Begin as Mattias responding to: *“Hi Mattias, welcome back. How have things been since last time?”*

The full prompt is available for testing in English and Swedish via shared conversation links. English version: https://chatgpt.com/share/69c19f9f-eb00-8000-8fb2-0bf0ae62e6e7 (accessed [22 August 2026]). Swedish version: https://chatgpt.com/share/69c1a00d-7bc0-8000-a49a-f92693bbb85e (accessed [22 August 2026]).

Appendix C — Student Survey



Leave survey

AI-generated Virtual Patients for Motivational Interviewing

This survey is part of a master's thesis on AI in psychology education and takes no more than 5 minutes to complete. Participation is voluntary and anonymous, and no personal data is collected.

1. Did you find the AI-tool useful for practicing MI?

- ○ Yes
- ○ No

2. How would you describe your experience using the AI tool? What did you like and what did you like about it when you compare it to real-play excercises carried out in class? What would you change about it?

3. Did the AI feel like a realistic patient during the interaction?

- ○ Yes
- ○ No

4. How comfortable did you feel interacting with the AI?

	Strongly disagree	Somewhat disagree	Neutral	Somewhat agree	Strongly agree
I felt more comfortable interacting with the AI virtual patient relationship than soing the real-play scenarios in class	○	○	○	○	○

5. Did you find the feedback provided by the AI useful?

- ○ Yes
- ○ No

6. What did you find most helpful (or unhelpful) about the feedback?

7. What kind of feedback did you prefer? (multiple selections are available)

- ☐ Qualitative feedback (short paragraph describing what I did right/wrong)
- ☐ Quantiative feedback (ammount of open ended questions/reflections/affirmations)
- ☐ Suggestions (what would have been a better response)
- ☐ None, I would rather not get any feedback
- ☐ I would prefer some other type of feedback

8. If you answered other on the previous question, please describe what type of feedback you would like to receive (Optional).

9. How would you prefer to receive feedback?

- ○ After each interaction with the virtual patient
- ○ Only once, at the end of the session

10. Would you use this tool again for practice?

- ○ Yes
- ○ No

11. What is your biggest concern or criticism of the tool? What is the biggest benefit you think this could provide for you?

Submit

Figure 1: Survey given to students